

POSTER PROPOSAL / ABSTRACT

Sobolev Inequalities on Warped Products

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Abstract

The Yamabe problem asks whether every conformal class of metrics on a closed Riemannian manifold contains a metric of constant scalar curvature. Beyond the existence problem, a natural question concerns the *quantitative stability of the Yamabe functional*, namely, how its energy deficit controls the distance to the set of minimizers. Although superquadratic stability is known to occur in certain settings, explicit examples remain scarce.

We study this question on warped products of the form $S^1 \times_f S^3$, where the metric on the S^3 -factor is scaled along S^1 by a positive warping function f . We consider an explicit family of warping functions giving rise to constant scalar curvature metrics. We also show that these warped products are not conformally equivalent to the round sphere S^4 , placing them outside the spherical conformal setting.

For the associated variational problem, we establish the existence and positivity of minimizers and show that they depend only on the S^1 -variable. This symmetry reduces the corresponding Euler-Lagrange equation to a nonlinear ordinary differential equation on S^1 . We are currently investigating conditions under which the only minimizers are constant functions, as a step toward understanding their degeneracy and constructing new examples of superquadratic stability for the Yamabe functional.